CS 6025: Intro to Parallel Processes Assignment 2

A) Design Summary:

1. Description of adaptive integration method

The adaptive integration method, specifically the **Adaptive Simpson's Rule**, is a technique for numerical integration that adjusts the size of intervals based on how much a function "wiggles" or changes in a specific region. Unlike standard methods that use a fixed step size, this approach focuses computational power only where it is needed to maintain accuracy. The process works by calculating a Simpson's Rule estimate for a given interval and comparing it against a refined estimate created by splitting that interval in half. If the difference between these two estimates—the error—is below the target tolerance, the result is accepted. If the error is too high, the algorithm recursively repeats the same logic on each half-interval. This strategy is highly efficient for irregular functions that are smooth in some areas but highly oscillatory in others, as it automatically concentrates more points in the difficult regions while using fewer points in the simpler ones.

2. Task granularity choice (K and rationale)

The task granularity is defined by the parameter **K**, which represents the number of equal-sized coarse intervals the domain [0, 1] is split into before the parallel computation begins. For this implementation, **K was set to 100** to ensure the number of tasks significantly exceeds the number of available processors (P). This choice provides a fine enough granularity to facilitate a balanced round-robin distribution in the static mode, preventing "straggler" processes where one rank remains busy with a complex region while others sit idle. To maintain numerical consistency and global accuracy, each of these 100 intervals is processed using a scaled local tolerance of tol/K, ensuring the final aggregated result meets the user's requested error bound.

3. Master/worker communication design

The Master/Worker communication design in Mode 2 utilizes a centralized coordination model built on MPI point-to-point primitives. The Master (Rank 0) acts as a task broker, maintaining a stack-based queue of intervals and responding to worker messages based on specific MPI tags, such as requests for work or the reporting of final integral results. The Workers (Ranks > 0) drive the computation by requesting intervals and performing the adaptive Simpson's error check. A key feature of this design is dynamic task generation: if an interval requires further subdivision, the worker processes one half locally and sends the other half back to the Master's queue. This "pull-based" approach ensures that highly oscillatory regions are distributed across all available processors, allowing idle workers to assist with high-computation areas discovered at runtime.

4. Termination strategy

The Termination Strategy in Mode 2 is designed to ensure the program only exits once all possible work is completed and accounted for, which is critical because the total number of sub-intervals is not known in advance. The strategy relies on the Master (Rank 0) tracking both the state of the task queue and the status of the workers; termination only occurs when the Master's task stack is empty and the count of "active" workers reaches zero. The Master increments an internal counter when dispatching an interval and decrements it only upon receiving a final result, ensuring that no worker is still in the middle of a calculation or about to generate a new sub-task. Once these two conditions are met, the Master initiates a

synchronized shutdown by sending a dedicated message with a termination tag to every worker rank. Upon receiving this signal, the workers break out of their request loops and proceed to the finalization phase. This coordinated approach prevents deadlocks or "orphan" processes and guarantees that the final global integral sum is complete and accurate before the MPI environment shuts down.

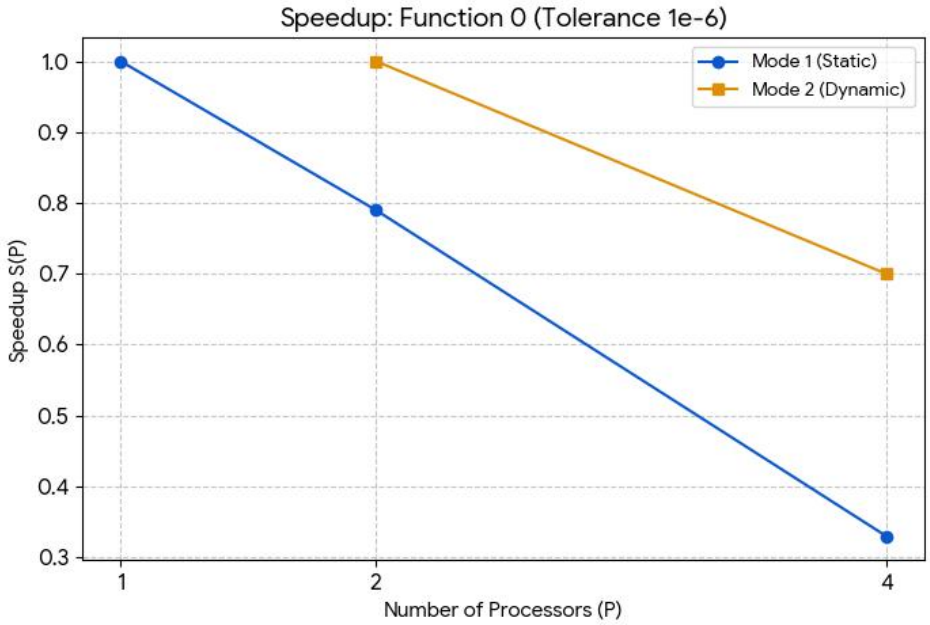
B) Result Tables:

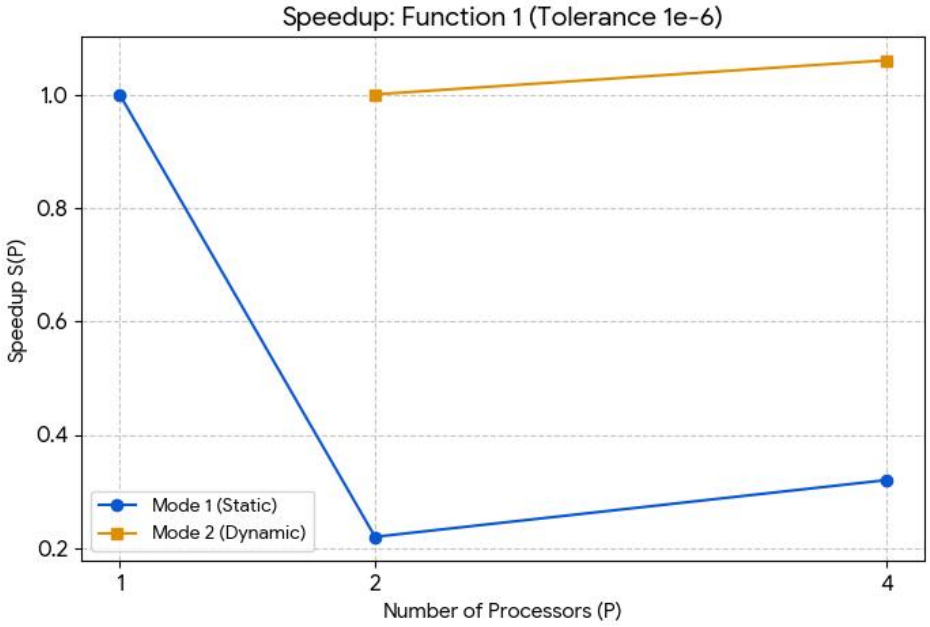
Added the excel file **A2.xlsx** along with this report in the ZIP file package.

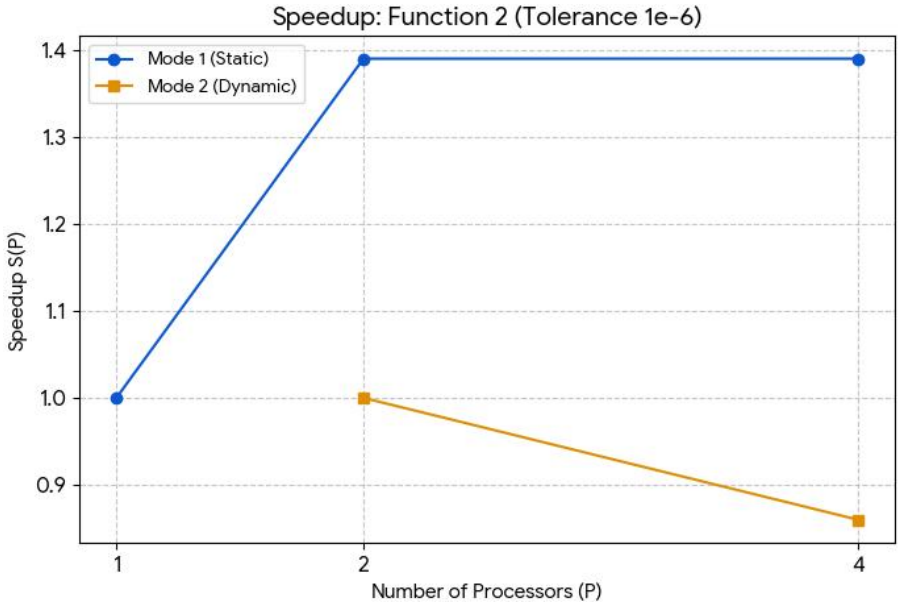
C) Plots:

1. Speedup for Function 0, 1, 2 - Comparison between Mode 1 vs Mode 2

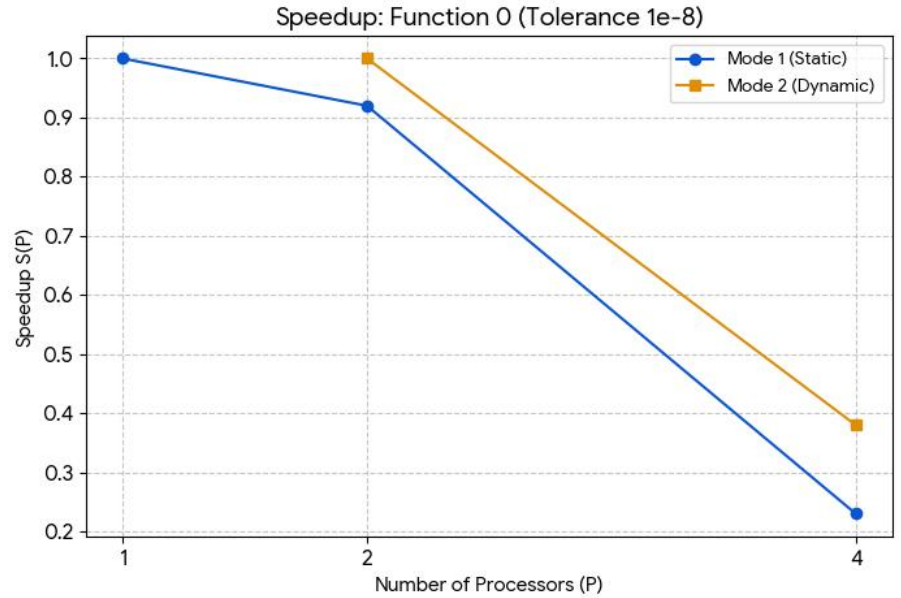
A) Tolerance 1e-6

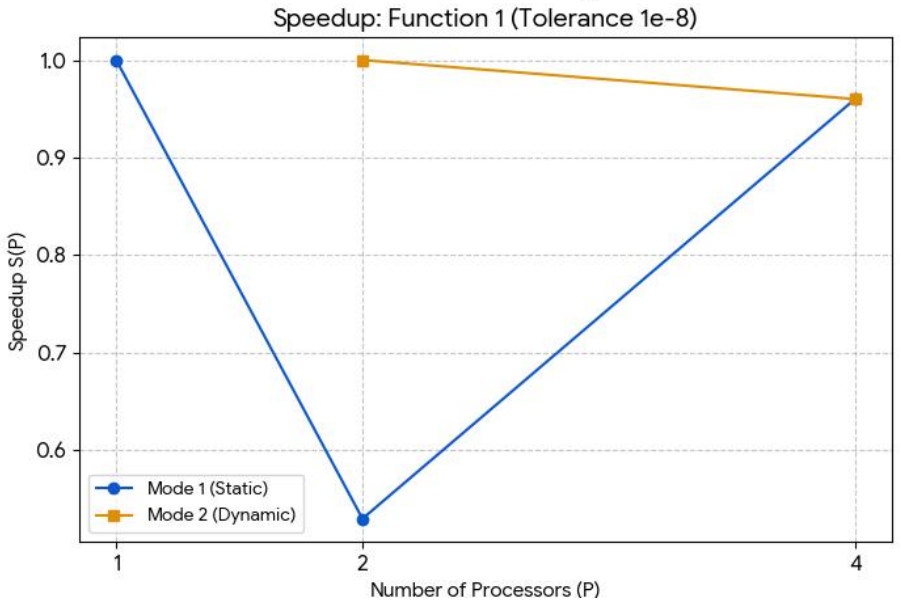


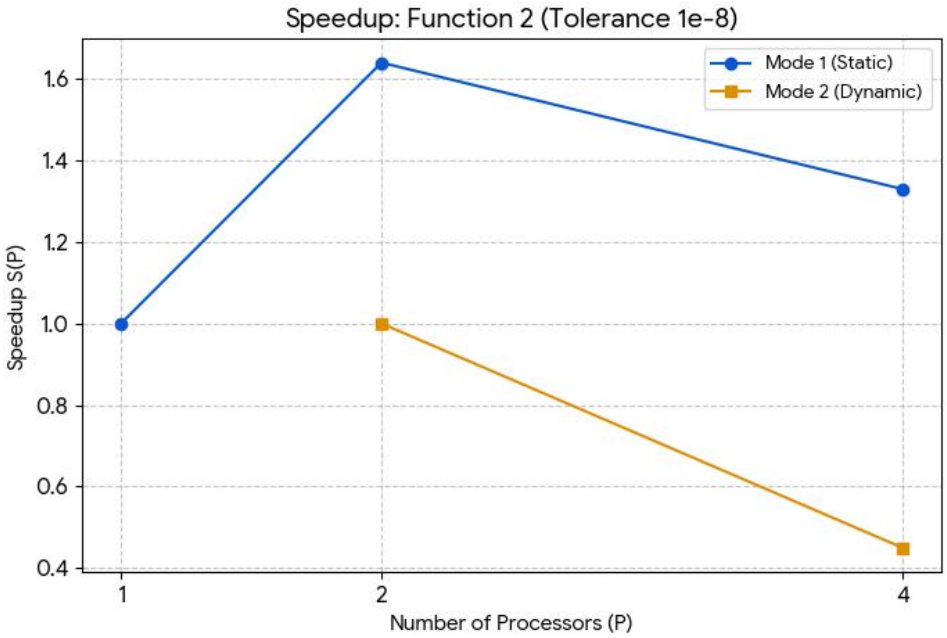




B) Tolerance 1e-8

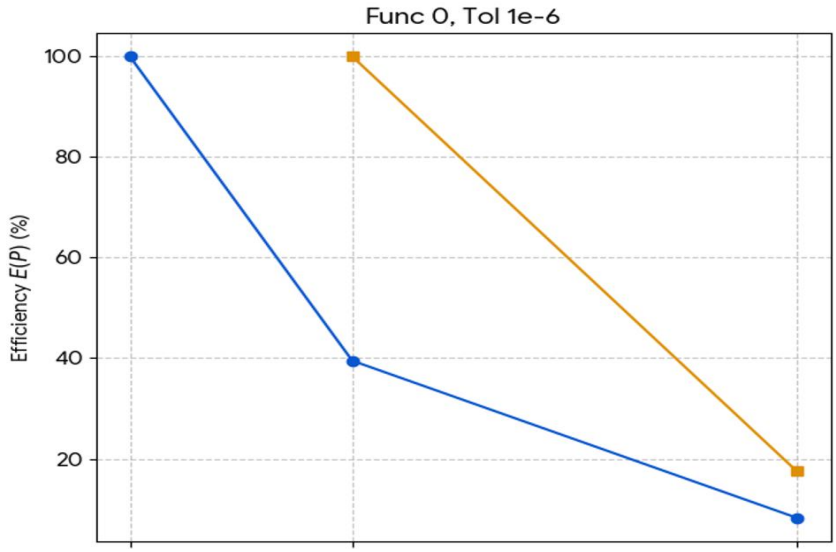


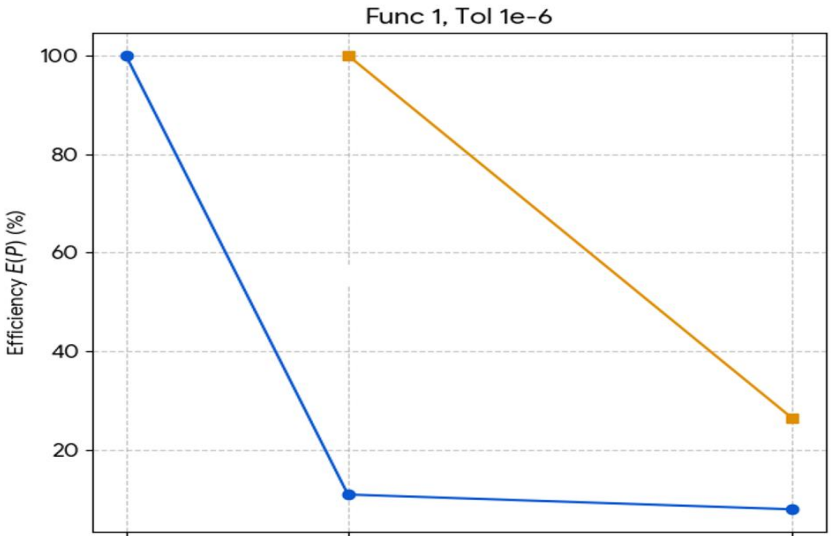


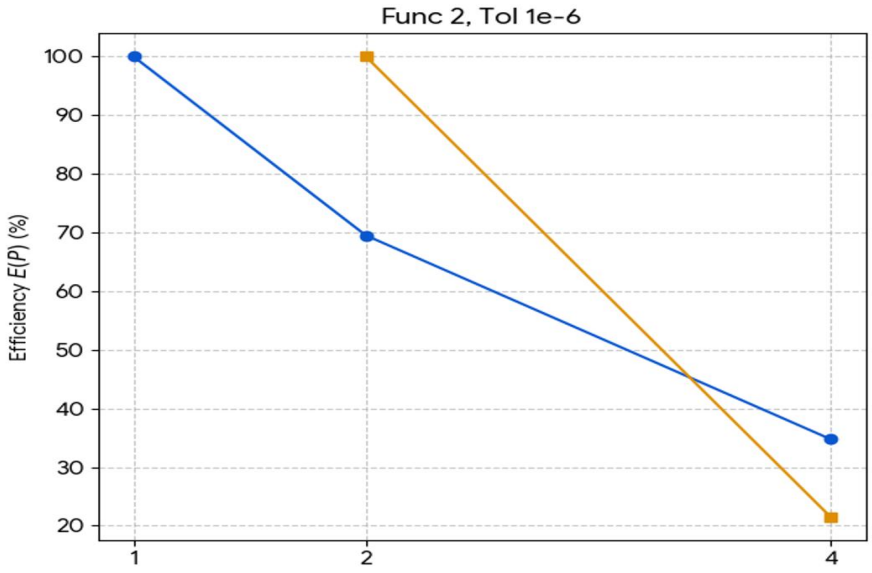


2. Efficiency for Function 0, 1, 2 - Comparison between Mode 1 vs Mode 2

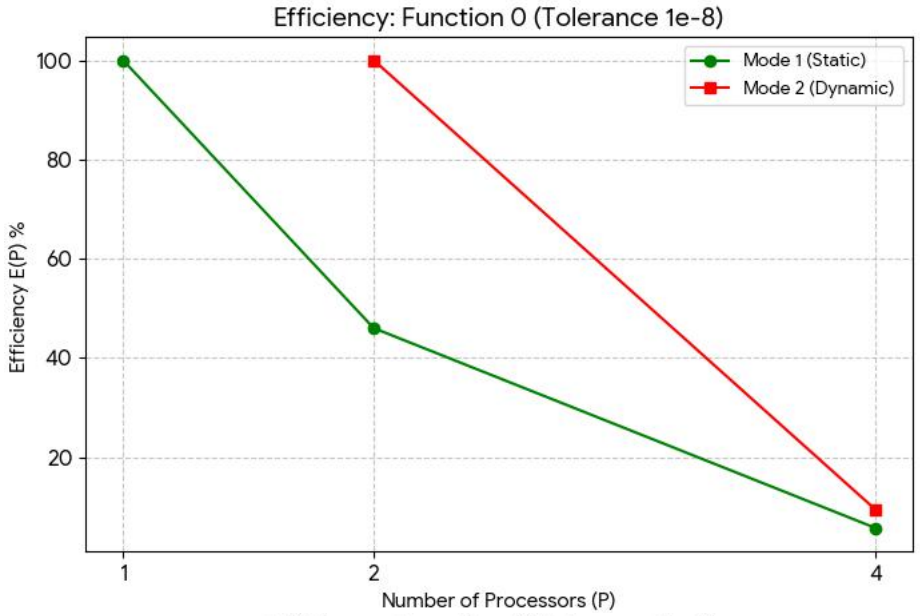
A) Tolerance 1e-6

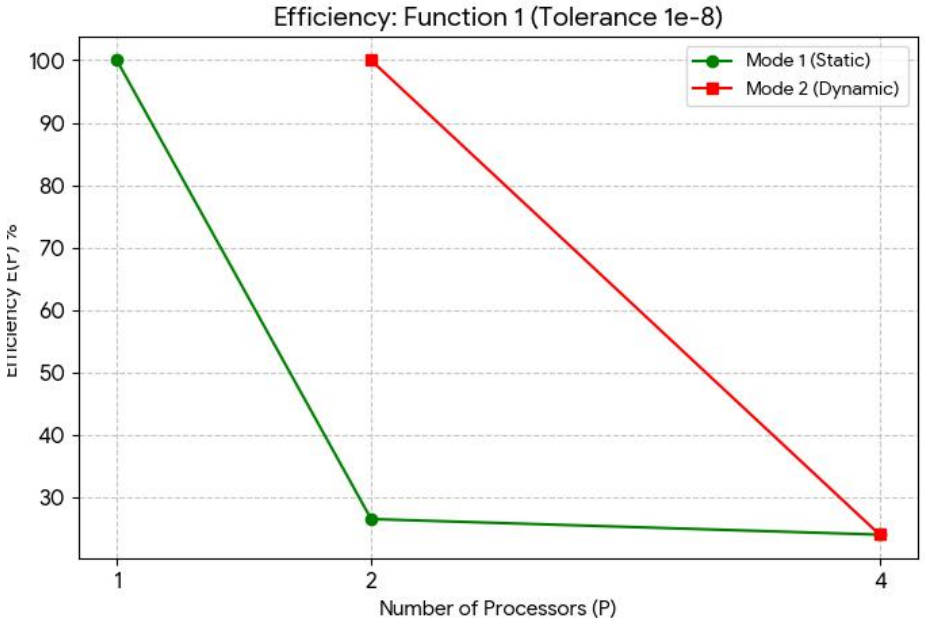


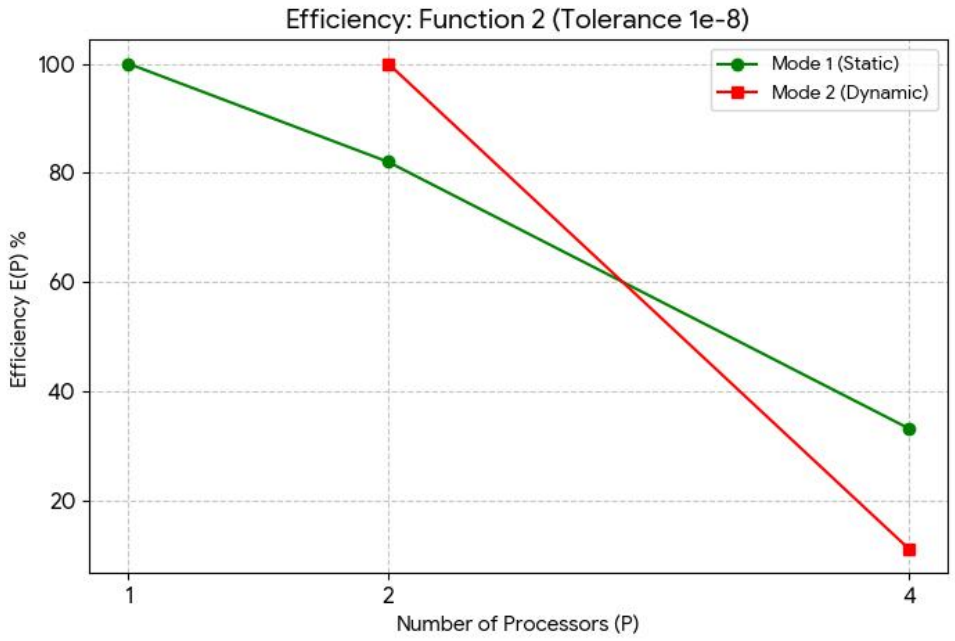




B) Tolerance 1e-8







D) Discussion:

1. Why does static decomposition perform poorly for adaptive workloads?

Static decomposition performs poorly for adaptive workloads because it relies on a fixed, pre-calculated distribution of tasks that cannot account for the unpredictable computational density of the function. In a static model like Mode 1, the domain is divided into equal-sized intervals before execution begins, assuming each part of the integral requires the same amount of effort. However, adaptive integration by nature concentrates more work in "difficult" regions where the function oscillates rapidly or has steep gradients. This leads to a significant load imbalance where some processors may finish their assigned intervals in milliseconds while others are stuck processing highly complex regions for much longer. Because there is no mechanism for idle processors to "steal" or receive new tasks from those that are overburdened, the total execution time is bottlenecked by the slowest processor. This inefficiency is reflected in the low speedup and efficiency values seen in your results, as the static approach fails to keep all available CPU cores busy throughout the entire computation.

2. How does dynamic task creation improve load balance?

Dynamic task creation improves load balance by allowing the workload to be redistributed in real-time as the "difficulty" of different regions is discovered. In adaptive integration, it is impossible to know beforehand which intervals will require deep recursion and which will be solved quickly. By enabling a worker to split an interval and send the extra half back to the master's queue, the system prevents a single processor from becoming a bottleneck while others sit idle. This "pull-based" approach ensures that as soon as a worker finishes its current task, it can immediately request a new one from the master. If one worker encounters a highly oscillatory region of the function that generates many sub-intervals, those new tasks are fed back into the central pool to be picked up by any available processor. This fluidity keeps the entire cluster busy throughout the computation, maximizing CPU utilization and significantly reducing the total execution time compared to a static distribution.

3. What is the cost of dynamic scheduling in MPI?

The cost of dynamic scheduling in MPI primarily stems from communication overhead and master-node bottlenecking. In a dynamic model, every task assigned requires at least two

messages: a request from the worker and a task dispatch from the master. When tasks are "fine-grained"—meaning the actual math takes very little time—the time spent packaging, sending, and receiving these MPI messages can actually exceed the time spent on the calculation itself. This is why your results show low efficiency for simple functions; the processors spend more time "talking" than "working." Furthermore, the Master-Worker bottleneck limits scalability because the Master (Rank 0) becomes a single point of congestion. As you add more processors, the Master must handle a rapidly increasing volume of requests, results, and new sub-tasks simultaneously. If the Master cannot keep up with the incoming messages, workers will sit idle waiting for their next assignment. Additionally, in many designs (like Mode 2), the Master node performs no actual integration, effectively "wasting" one entire CPU core's worth of computational power, which inherently lowers the overall efficiency of the system as the total processor count increases.

4. Which function benefited most from dynamic scheduling, and why?

Function 2 (f(x) = \sin(200x)e^{-x}) benefited the most from dynamic scheduling because it possesses the highest degree of "load imbalance" or irregularity. Unlike the other functions, Function 2 is highly oscillatory in specific regions and smoother in others, meaning the computational effort required is not distributed evenly across the domain [0, 1]. In a static decomposition, a processor assigned to a dense, oscillatory patch would be overwhelmed with recursive subdivisions, while a processor assigned to a smooth patch would finish almost instantly and sit idle.Dynamic scheduling addresses this by allowing the worker assigned to the "heavy" oscillatory region to split its intervals and send them back to the Master's queue. This mechanism enables idle workers—who may have finished their "easy" smooth sections—to pick up the extra work and assist in the computation. By constantly redistributing these bursts of tasks, the dynamic mode ensures that all available processors remain productive, resulting in more consistent speedup and better hardware utilization for complex, unpredictable functions.

E) Sample Outputs for each mode and function:

1. Processors - 2; Tolerance level - 1e+6; Functions - 0,1,2

```
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 2 ./integration 0 2 1e-6

--- Results (Mode 2) ---
Integral: 0.4832176932
Total Intervals: 4
Function ID: 0
Error Tolerance: 0.0000010000
Execution Time: 0.000053 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 2 ./integration 0 1 1e-6

--- Results (Mode 1) ---
Integral: 0.4832176955
Total Intervals: 100
Function ID: 0
Error Tolerance: 0.0000010000
Execution Time: 0.000040 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 2 ./integration 0 0 1e-6

--- Results (Mode 0) ---
Integral: 0.4832176988
Total Intervals: 7
Function ID: 0
Error Tolerance: 0.0000010000
Execution Time: 0.000019 seconds
```

```
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 2 ./integration 1 2 1e-6

--- Results (Mode 2) ---
Integral: 0.2677945770
Total Intervals: 14
Function ID: 1
Error Tolerance: 0.0000010000
Execution Time: 0.000084 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 2 ./integration 1 1 1e-6

--- Results (Mode 1) ---
Integral: 0.2677945045
Total Intervals: 100
Function ID: 1
Error Tolerance: 0.0000010000
Execution Time: 0.000080 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 2 ./integration 1 0 1e-6

--- Results (Mode 0) ---
Integral: 0.2677944875
Total Intervals: 22
Function ID: 1
Error Tolerance: 0.0000010000
Execution Time: 0.000028 seconds
```

```
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 2 ./integration 2 2 1e-6

--- Results (Mode 2) ---
Integral: -0.2586536266
Total Intervals: 2
Function ID: 2
Error Tolerance: 0.0000010000
Execution Time: 0.000045 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 2 ./integration 2 1 1e-6

--- Results (Mode 1) ---
Integral: 0.0041117972
Total Intervals: 427
Function ID: 2
Error Tolerance: 0.0000010000
Execution Time: 0.000112 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 2 ./integration 2 0 1e-6

--- Results (Mode 0) ---
Integral: -0.2586536118
Total Intervals: 3
Function ID: 2
Error Tolerance: 0.0000010000
Execution Time: 0.000016 seconds
```

2. Processors - 4; Tolerance level - 1e+6; Functions - 0,1,2

```
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 4 ./integration 0 2 1e-6

--- Results (Mode 2) ---
Integral: 0.4832176932
Total Intervals: 4
Function ID: 0
Error Tolerance: 0.0000010000
Execution Time: 0.000079 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 4 ./integration 0 1 1e-6

--- Results (Mode 1) ---
Integral: 0.4832176955
Total Intervals: 100
Function ID: 0
Error Tolerance: 0.0000010000
Execution Time: 0.000049 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 4 ./integration 0 0 1e-6

--- Results (Mode 0) ---
Integral: 0.4832176988
Total Intervals: 7
Function ID: 0
Error Tolerance: 0.0000010000
Execution Time: 0.000036 seconds
```

```
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 4 ./integration 1 2 1e-6

--- Results (Mode 2) ---
Integral: 0.2677945770
Total Intervals: 14
Function ID: 1
Error Tolerance: 0.0000010000
Execution Time: 0.000087 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 4 ./integration 1 1 1e-6

--- Results (Mode 1) ---
Integral: 0.2677945045
Total Intervals: 100
Function ID: 1
Error Tolerance: 0.0000010000
Execution Time: 0.000175 seconds
^[[Amvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 4 ./integration 1 0 1e-6

--- Results (Mode 0) ---
Integral: 0.2677944875
Total Intervals: 22
Function ID: 1
Error Tolerance: 0.0000010000
Execution Time: 0.000033 seconds
```

```
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 4 ./integration 2 2 1e-6

--- Results (Mode 2) ---
Integral: -0.2586536266
Total Intervals: 2
Function ID: 2
Error Tolerance: 0.0000010000
Execution Time: 0.000044 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 4 ./integration 2 1 1e-6

--- Results (Mode 1) ---
Integral: 0.0041117972
Total Intervals: 427
Function ID: 2
Error Tolerance: 0.0000010000
Execution Time: 0.000360 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 4 ./integration 2 0 1e-6

--- Results (Mode 0) ---
Integral: -0.2586536118
Total Intervals: 3
Function ID: 2
Error Tolerance: 0.0000010000
Execution Time: 0.000042 seconds
```

3. Processors - 2; Tolerance level - 1e+8; Functions - 0,1,2

```
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 2 ./integration 0 2 1e-8

--- Results (Mode 2) ---
Integral: 0.4832176955
Total Intervals: 14
Function ID: 0
Error Tolerance: 0.0000000100
Execution Time: 0.000146 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 2 ./integration 0 1 1e-8

--- Results (Mode 1) ---
Integral: 0.4832176955
Total Intervals: 100
Function ID: 0
Error Tolerance: 0.0000000100
Execution Time: 0.000081 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 2 ./integration 0 0 1e-8

--- Results (Mode 0) ---
Integral: 0.4832176955
Total Intervals: 23
Function ID: 0
Error Tolerance: 0.0000000100
Execution Time: 0.000026 seconds
```

```
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 2 ./integration 1 2 1e-8

--- Results (Mode 2) ---
Integral: 0.2677945083
Total Intervals: 32
Function ID: 1
Error Tolerance: 0.0000000100
Execution Time: 0.000152 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 2 ./integration 1 1 1e-8

--- Results (Mode 1) ---
Integral: 0.2677945045
Total Intervals: 116
Function ID: 1
Error Tolerance: 0.0000000100
Execution Time: 0.000071 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 2 ./integration 1 0 1e-8

--- Results (Mode 0) ---
Integral: 0.2677945045
Total Intervals: 71
Function ID: 1
Error Tolerance: 0.0000000100
Execution Time: 0.000079 seconds
```

```
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 2 ./integration 2 2 1e-8

--- Results (Mode 2) ---
Integral: -0.2586535817
Total Intervals: 7
Function ID: 2
Error Tolerance: 0.0000000100
Execution Time: 0.000244 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 2 ./integration 2 1 1e-8

--- Results (Mode 1) ---
Integral: 0.0041117973
Total Intervals: 1492
Function ID: 2
Error Tolerance: 0.0000000100
Execution Time: 0.000382 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 2 ./integration 2 0 1e-8

--- Results (Mode 0) ---
Integral: -0.2586535817
Total Intervals: 7
Function ID: 2
Error Tolerance: 0.0000000100
Execution Time: 0.000021 seconds
```

4. Processors - 4; Tolerance level - 1e+8; Functions - 0,1,2

```
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 4 ./integration 0 2 1e-8

--- Results (Mode 2) ---
Integral: 0.4832176955
Total Intervals: 14
Function ID: 0
Error Tolerance: 0.0000000100
Execution Time: 0.001078 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 4 ./integration 0 1 1e-8

--- Results (Mode 1) ---
Integral: 0.4832176955
Total Intervals: 100
Function ID: 0
Error Tolerance: 0.0000000100
Execution Time: 0.000218 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 4 ./integration 0 0 1e-8

--- Results (Mode 0) ---
Integral: 0.4832176955
Total Intervals: 23
Function ID: 0
Error Tolerance: 0.0000000100
Execution Time: 0.000042 seconds
```

```
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 4 ./integration 1 2 1e-8

--- Results (Mode 2) ---
Integral: 0.2677945083
Total Intervals: 32
Function ID: 1
Error Tolerance: 0.0000000100
Execution Time: 0.000167 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 4 ./integration 1 1 1e-8

--- Results (Mode 1) ---
Integral: 0.2677945045
Total Intervals: 116
Function ID: 1
Error Tolerance: 0.0000000100
Execution Time: 0.000251 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 4 ./integration 1 0 1e-8

--- Results (Mode 0) ---
Integral: 0.2677945045
Total Intervals: 71
Function ID: 1
Error Tolerance: 0.0000000100
Execution Time: 0.000052 seconds
```

```
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 4 ./integration 2 2 1e-8

--- Results (Mode 2) ---
Integral: -0.2586535817
Total Intervals: 7
Function ID: 2
Error Tolerance: 0.0000000100
Execution Time: 0.000107 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 4 ./integration 2 1 1e-8

--- Results (Mode 1) ---
Integral: 0.0041117973
Total Intervals: 1492
Function ID: 2
Error Tolerance: 0.0000000100
Execution Time: 0.000364 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 4 ./integration 2 0 1e-8

--- Results (Mode 0) ---
Integral: -0.2586535817
Total Intervals: 7
Function ID: 2
Error Tolerance: 0.0000000100
Execution Time: 0.000028 seconds
```

5. Processors - 1; Tolerance level - 1e+6; Functions - 0,1

```
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 1 ./integration 0 0 1e-6

--- Results (Mode 0) ---
Integral: 0.4832176988
Total Intervals: 7
Function ID: 0
Error Tolerance: 0.0000010000
Execution Time: 0.000021 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 1 ./integration 0 1 1e-6

--- Results (Mode 1) ---
Integral: 0.4832176955
Total Intervals: 100
Function ID: 0
Error Tolerance: 0.0000010000
Execution Time: 0.000037 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 1 ./integration 0 2 1e-6


^C
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$
```

```
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 1 ./integration 1 0 1e-6

--- Results (Mode 0) ---
Integral: 0.2677944875
Total Intervals: 22
Function ID: 1
Error Tolerance: 0.0000010000
Execution Time: 0.000028 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 1 ./integration 1 1 1e-6

--- Results (Mode 1) ---
Integral: 0.2677945045
Total Intervals: 100
Function ID: 1
Error Tolerance: 0.0000010000
Execution Time: 0.000044 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 1 ./integration 1 2 1e-6


^C
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$
```

```
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 1 ./integration 2 0 1e-6

--- Results (Mode 0) ---
Integral: -0.2586536118
Total Intervals: 3
Function ID: 2
Error Tolerance: 0.0000010000
Execution Time: 0.000016 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 1 ./integration 2 1 1e-6

--- Results (Mode 1) ---
Integral: 0.0041117972
Total Intervals: 427
Function ID: 2
Error Tolerance: 0.0000010000
Execution Time: 0.000193 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 1 ./integration 2 2 1e-6


^C
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$
```

6. Processors - 1; Tolerance level - 1e+8; Functions - 0,1

```
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 1 ./integration 0 0 1e-8

--- Results (Mode 0) ---
Integral: 0.4832176955
Total Intervals: 23
Function ID: 0
Error Tolerance: 0.0000000100
Execution Time: 0.000025 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 1 ./integration 0 1 1e-8

--- Results (Mode 1) ---
Integral: 0.4832176955
Total Intervals: 100
Function ID: 0
Error Tolerance: 0.0000000100
Execution Time: 0.000053 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 1 ./integration 0 2 1e-8

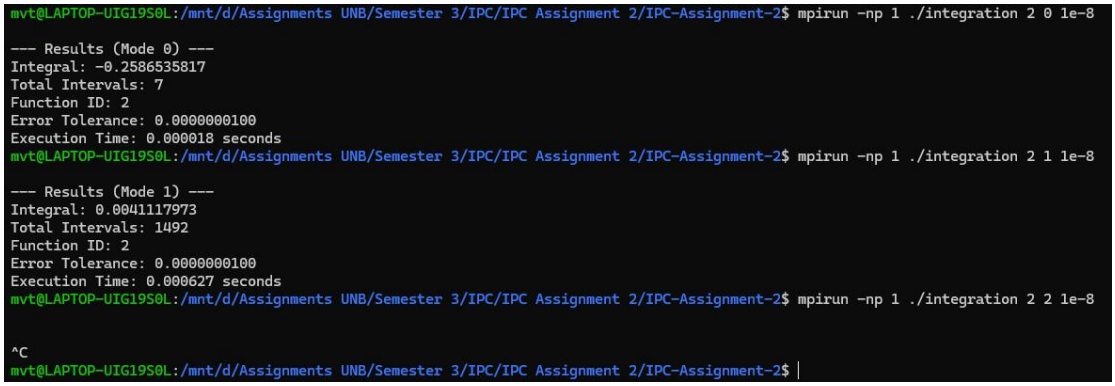

^C
```

```
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 1 ./integration 1 0 1e-8

--- Results (Mode 0) ---
Integral: 0.2677945045
Total Intervals: 71
Function ID: 1
Error Tolerance: 0.0000000100
Execution Time: 0.000052 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 1 ./integration 1 1 1e-8

--- Results (Mode 1) ---
Integral: 0.2677945045
Total Intervals: 116
Function ID: 1
Error Tolerance: 0.0000000100
Execution Time: 0.000049 seconds
mvt@LAPTOP-UIG19S0L:/mnt/d/Assignments UNB/Semester 3/IPC/IPC Assignment 2/IPC-Assignment-2$ mpirun -np 1 ./integration 1 2 1e-8


^C
```



7. Processors - 1; Mode 2; all functions and fault tolerance

Faced infinite runtime due to deadlock. Only one processor accesses the master and since there are no processors to accesses the worker block, the master waits infinitely for the worker. In the above screenshots for output with 1 processor, I have executed for mode 2, but after infinite loop, so I terminated the runtime.